MUHAMMAD ALI

Full-Stack, Mobile & Computer Vision Systems Engineer

muhammadali.chd12@gmail.com
+92 318 4439061
Peshawar, KPK, Pakistan
github.com/Muhammad-ali234
linkedin.com/in/muhammad-ali

TECHNICAL SKILLS

Programming Languages
JavaScript (ES6+), TypeScript, Python, Dart, C++, SQL

Frontend & Web
React.js, Tailwind CSS, HTML5, CSS3, Vite, Component Architecture, Responsive UI

Mobile Engineering
Flutter, Dart, Provider, State Architecture, Hive, Material 3, SQLite, Device Hardware APIs

Backend & Databases
Node.js, Express.js, FastAPI, MySQL, PostgreSQL, MongoDB, Firebase, REST APIs, WebSockets

AI & Computer Vision
YOLOv8, OpenCV, ByteTrack, Multi-Object Tracking, Real-Time Edge Inference, Plant Telemetry

Developer Tools & Printing
Git, GitHub, Postman, JWT Auth, Thermal Printing (ESC/POS 80mm/58mm), A4/A5 PDF Engine

EDUCATION

Bachelor of Science in Software Engineering (BSSE)
City University of Science & IT, Peshawar
2020 – 2024 | Peshawar, Pakistan

CERTIFICATIONS

Flutter Application Development
Coursera • May 2025
Verify Credential ↗

Web Design Specialization
Coursera • Feb 2023
Verify Credential ↗

Flutter Fellowship Graduate
Bytewise Limited • Jun 2023
Verify Credential ↗

KEY STRENGTHS

- Enterprise POS & Accounting
- Production Computer Vision
- 4-Role Mobile Architectures
- Database Normalization & ACID
- End-to-End Client Deployments

PROFESSIONAL SUMMARY

Results-driven Software Engineer specialized in architecting and delivering full-stack web applications (React.js, Node.js), high-performance cross-platform mobile apps (Flutter), and industrial computer vision systems (YOLOv8, OpenCV). Proven track record of taking complex commercial client requirements from initial system design to production deployment, including commercial Accounting & POS software delivered to 10+ shops, factory ERP & pipe automation delivered to A-One PVC Company, and multi-party automotive service platforms.

DELIVERED FLAGSHIP PROJECTS

Accounting & POS Software — Enterprise ERP

Delivered to 10+ Shop Clients

React.js • Node.js • Express • MySQL • Tailwind CSS • WebSockets • JWT

- Engineered a full-featured commercial financial platform featuring rapid POS checkout, supplier purchases, and bidirectional returns (sales & purchase returns) with strict transactional balance recalculations.
- Implemented customer & vendor ledgers, quotation-to-sales conversion pipeline, operational expense categorization, and bank account / cheque clearance reconciliation workflows.
- Built granular inventory stock valuation, GST tax reports, staff audit trail logs, and dynamic printing engine supporting thermal receipts (80mm/58mm), A4, and A5 invoice layouts.

PipeVision AI — Factory ERP & Automated Pipe Counting

Delivered to A-One PVC Company

Python • YOLOv8 • OpenCV • React.js • Node.js • FastAPI • MySQL • WebSockets

- Delivered an integrated dual-engine manufacturing platform combining comprehensive factory ERP operations with real-time industrial conveyor computer vision automation.
- Built factory ERP modules covering biometric & face shift attendance for 60+ workers, daily sales dispatches & truck gate passes, supplier resin purchases (tons), and factory OPEX ledgers.
- Trained custom YOLOv8 model for 60 FPS real-time conveyor tracking, automated pipe tallying (>14,800 units/shift), diameter anomaly detection, and sub-millisecond WebSocket plant floor telemetry.

VADS — Vehicle At Doorstep Platform

Delivered to 1 Automotive Client

Flutter • Node.js • Express • MySQL • Socket.io • Provider • Firebase Auth

- Architected a multi-role automotive platform connecting buyers, sellers, drivers, and employees with real-time Socket.io chat, dynamic doorstep test drive calendar bookings, and vehicle inspections.

WORK EXPERIENCE

Full-Stack & Mobile Software Engineer

Aug 2024 – Present

Independent Client Deployments & Freelance Engineering | Peshawar, Pakistan

- Successfully architected, coded, and deployed custom software solutions across retail, manufacturing, and automotive domains, managing database schemas, backend APIs, and frontends.

Flutter Mobile Developer Intern

Apr 2024 – Jul 2024

FF-Steel | Peshawar, Pakistan

- Engineered mobile UI components in Flutter, connected REST APIs, and managed application state using Provider.

Flutter Mobile Fellow

Mar 2023 – Jun 2023

Bytewise Limited | Remote

- Completed intensive hands-on fellowship covering mobile clean architecture, custom widgets, and collaborative Git practices.